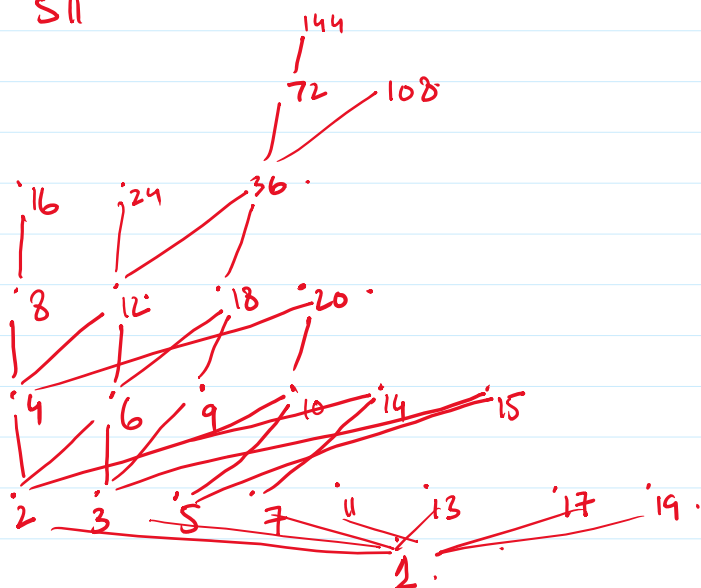


lecture 18.

Ex 20 :-
S11

$(\mathbb{Z}^+, 1)$.



$\text{GLB}(\{3, 9, 12\})$.
 $\text{LUB}(\{3, 9, 12\})$.

$\text{UB}(\{3, 9, 12\}) = \{36, 72, 108, 144, \dots\}$.

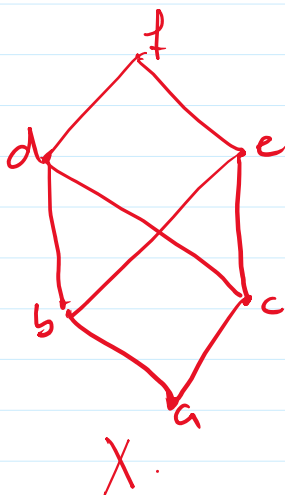
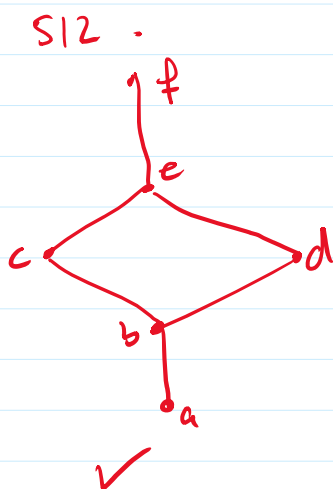
$\text{LUB}(\{3, 9, 12\}) = 36$.
 $= \text{LCM}$.

$\text{LUB}(\{3, 9, 12\}) = \{3, 1\}$.
 $\text{GLB}(\{3, 9, 12\}) = 3$.
 $\hookrightarrow \text{CD}$.

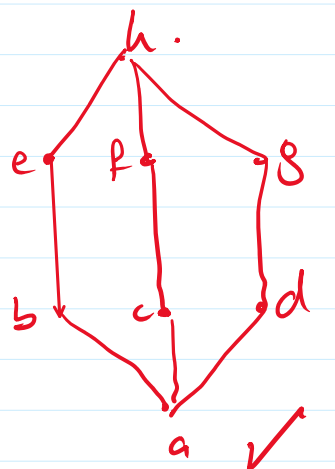
LATICE. $\forall a, b \in S$

$(S, \leq)$.

$\text{GLB}(\{a, b\})$ exists $\wedge$ $\text{LUB}(\{a, b\})$ exists.



$\text{GLB}(\{b, c\})$ does not exist.



Graphs. $G = (V, E)$.

$V =$ Set of Vertices.
 $E =$ " " Edges.

Undirected Graphs.

1- Simple. - No loops -
 - No multiedges -

2- Multigraph. Contain multiedges.

3- Pseudograph. loop + may contain multiedges -

directed Graphs.

1- Simple. - No loops -
 - No multiedges -

2- Multigraph. Contain multiedges.

3- Pseudograph. loop + may contain multiedges -

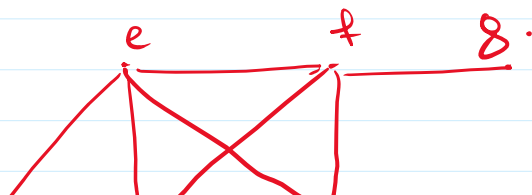
mixed Graph. Directed + Undirected.

Null Graph. (No Vertex).

Finite + Infinite Graphs.

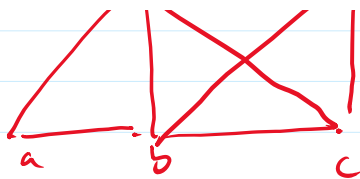
Degree of a Graph. (Undirected).

Ex 2
556



$deg(a) = 2$
 $deg(b) = 4$
...

16



d.

$V = \{a, b, c, d, e, f, g\}$.

17

$$d(b) = 4$$

$$d(c) = 3$$

$$d(d) = 0$$

$$d(e) = 4$$

$$d(f) = 4$$

$$d(g) = 1$$

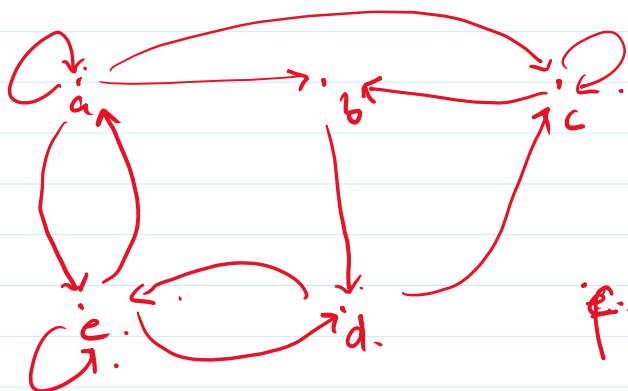
$$\sum_{u \in V} d(u) = 18$$

Handshaking.

$$2|E| = \sum_{u \in V} d(u).$$

Directed Graphs (Degree).

Ex 4 :-
538



$$deg^-(a) = 2$$

$$deg^-(b) = 2$$

$$deg^-(c) = 3$$

$$deg^-(d) = 2$$

$$deg^-(e) = 3$$

$$deg^-(f) = 0$$

$$\sum_{u \in V} deg^-(u) = 12$$

$$deg^+(a) = 4$$

$$deg^+(b) = 1$$

$$deg^+(c) = 2$$

$$deg^+(d) = 2$$

$$deg^+(e) = 3$$

$$deg^+(f) = 0$$

$$\sum_{u \in V} deg^+(u) = 12$$

$$|E| = \sum_{u \in V} deg^-(u) = \sum_{u \in V} deg^+(u).$$

Special types of Simple Graphs

1 - Complete Graph. K_n $n = \text{number of vertices.}$

1 - Complete Graph. K_n $n = \text{number of vertices.}$

K_1

K_2

K_3

K_4

K_5

K_6

2 - Cycles. C_n $n \geq 3$ $n = \text{number of vertices.}$

C_3

C_4

C_5

C_6

3 - Wheels. W_n $n \geq 3$ $n = \text{number of vertices.}$

W_3

W_4

W_5

W_6

4 - Quboid. Q_n

Q_1

Q_2

Q_3

0 2.
Q1.

00 02.
Q2.

000 001.
Q3.

Vertices		Edges -	
K_1	1		0
K_2	2		1
K_3	3		3
K_4	4		6
K_5	5		10
	1		15
	⋮		⋮
K_n	n .		$\frac{n(n-1)}{2}$.
C_3	3		3
C_4	4		4
C_5	5		5
	⋮		⋮
C_n	n		n .
W_3	4		6
W_4	5		8
W_5	6		10
	⋮		⋮
W_n	$n+1$.		$2n$.
Q_1	2		1
Q_2	4		4
Q_3	8		12
	⋮		⋮
Q_n	2^n		2^n